

Hand-Drawn Chemical Structure Recognition

Does synthetic drawing-style augmentation improve real-image performance?

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BT5230 Mini Project – Challenge 72

Project-trained MolScribe comparison with an independently labelled DECIMER reference

Problem and project objective

Optical chemical structure recognition (OCSR) converts a molecular depiction into a machine-readable representation such as SMILES.

Hand-drawn structures contain uneven bonds, distorted rings, variable spacing, and ambiguous stereochemical marks. These variations create a domain shift from clean computer-rendered images.

Source: Rajan et al., Journal of Cheminformatics 16, 2024, article 78.

Challenge 72 objective

Test whether matched synthetic hand-drawn-style augmentation improves recognition of unseen real drawings under student-scale compute.

Output: predicted SMILES, checked for chemical validity and structural agreement.

Data and leakage controls

ChEMBL-derived development set

- 3,758 training molecules
- 470 clean validation molecules
- 470 held-out synthetic test molecules
- Scaffold-separated partitions

Real benchmark

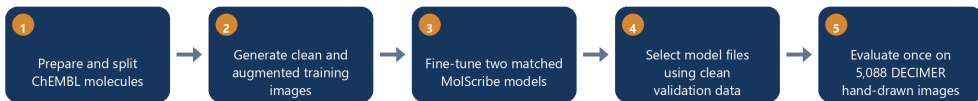
- 5,088 DECIMER hand-drawn images
- DECIMER test images were not used for training or model selection
- Every row retained in the denominator
- No retuning after benchmark evaluation

Matched comparison and split provenance

Clean and augmented conditions used identical molecule IDs, labels, starting checkpoint, validation images, seed, optimizer settings, and one-epoch duration. The pipeline used the master manifest; separate split CSVs are unchanged inspection exports.

Dataset: DECIMER Hand-Drawn Molecule Images v1.2, CC BY 4.0.

Experimental workflow



Leakage control: DECIMER test images were used only for the final comparison.

Independent variable: training depiction style. **Primary endpoint:** exact isomeric match on real drawings.

Model and evaluation metrics

Model

- Pretrained MolScribe OCSR model
- Image encoder and molecular sequence decoder
- Clean and augmented one-epoch fine-tuning
- Model files selected using clean validation only

Chemistry-aware metrics

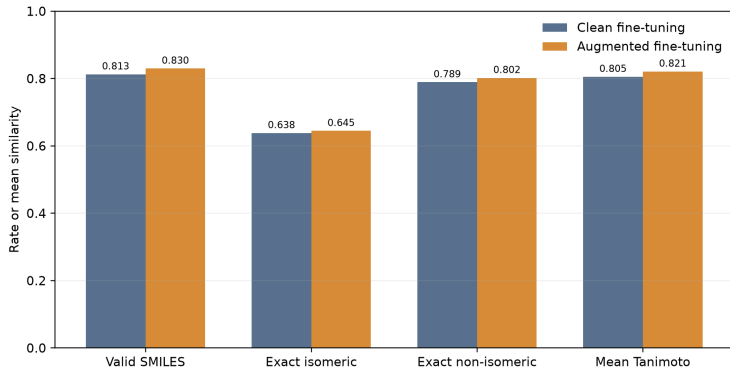
- Valid SMILES after RDKit sanitization
- Exact isomeric match with stereochemistry
- Exact non-isomeric match without stereochemistry
- Morgan-fingerprint Tanimoto similarity

Conservative scoring

Invalid predictions received zero exact-match and zero similarity scores. Paired bootstrap intervals quantified uncertainty between the two checkpoints.

Checkpoint SHA-256 identifiers and public view-only download links are documented in the submitted README.

Synthetic validation showed small gains



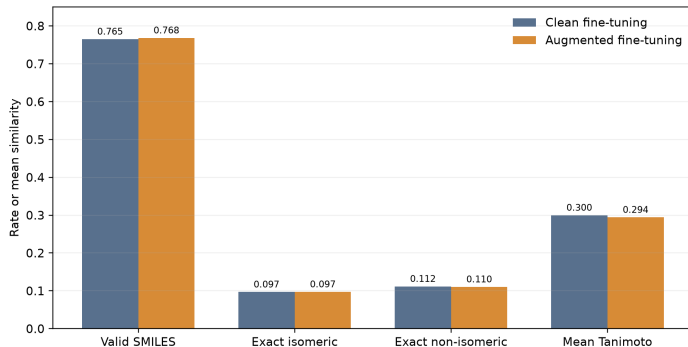
470 clean images

Exact isomeric:
63.83% to 64.47%

Mean Tanimoto:
0.8054 to 0.8210

This was a feasibility
check, not the final
real-image result.

Augmentation did not improve the held-out real benchmark



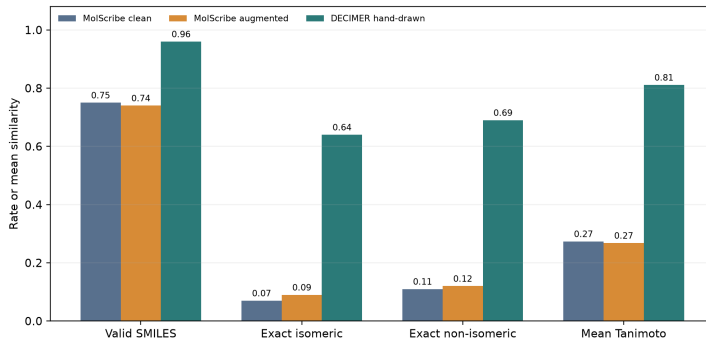
All 5,088 images

Exact isomeric:
9.69% vs. 9.73%
 $\Delta = +0.04$ pp

Mean Tanimoto:
0.2998 vs. 0.2941
 $\Delta = -0.0057$
95% CI: $[-0.0111, -0.0005]$

Conclusion: the tested augmentation did not improve real-image recognition under this design.

External DECIMER reference shows attainable performance



Fixed seeded sample
100 benchmark images

Official DECIMER model:
64.00% exact isomeric
0.8111 mean Tanimoto

External model only. Its
weights were not trained or
modified in this project.

DECIMER exact-match Wilson 95% CI: 54.24% to 72.73%. The sample result is not a full-benchmark estimate.

Limitations and research integrity

Observed limitations

- Accuracy declined for larger, ring-rich, and stereochemically complex structures.
- One epoch and 3,758 training molecules limited adaptation.
- Synthetic drawing styles did not reproduce the full real-image distribution.
- Valid SMILES did not guarantee the correct molecule.

Integrity safeguards

- Molecules were split before rendering.
- DECIMER test images were used only for the final comparison.
- Invalid outputs remained in the denominator.
- DECIMER results were labelled as an external reference.
- AI tools assisted coding and documentation. The student verified the analysis and remains responsible for all claims.

Responsible use

This prototype supports research and digitization workflows. Predictions require expert verification before scientific use.

Conclusions

1. The experiment delivered a leakage-controlled OCSR comparison on all 5,088 real hand-drawn benchmark images.
2. The tested synthetic augmentation produced small clean-validation gains but did not improve real-image recognition.
3. The external DECIMER result indicates that larger, domain-specific training can achieve substantially better hand-drawn recognition.

Next research step

Test stronger drawing-style generators and longer transfer schedules using an independent development set, then evaluate once on a newly held-out test set.

Thank you